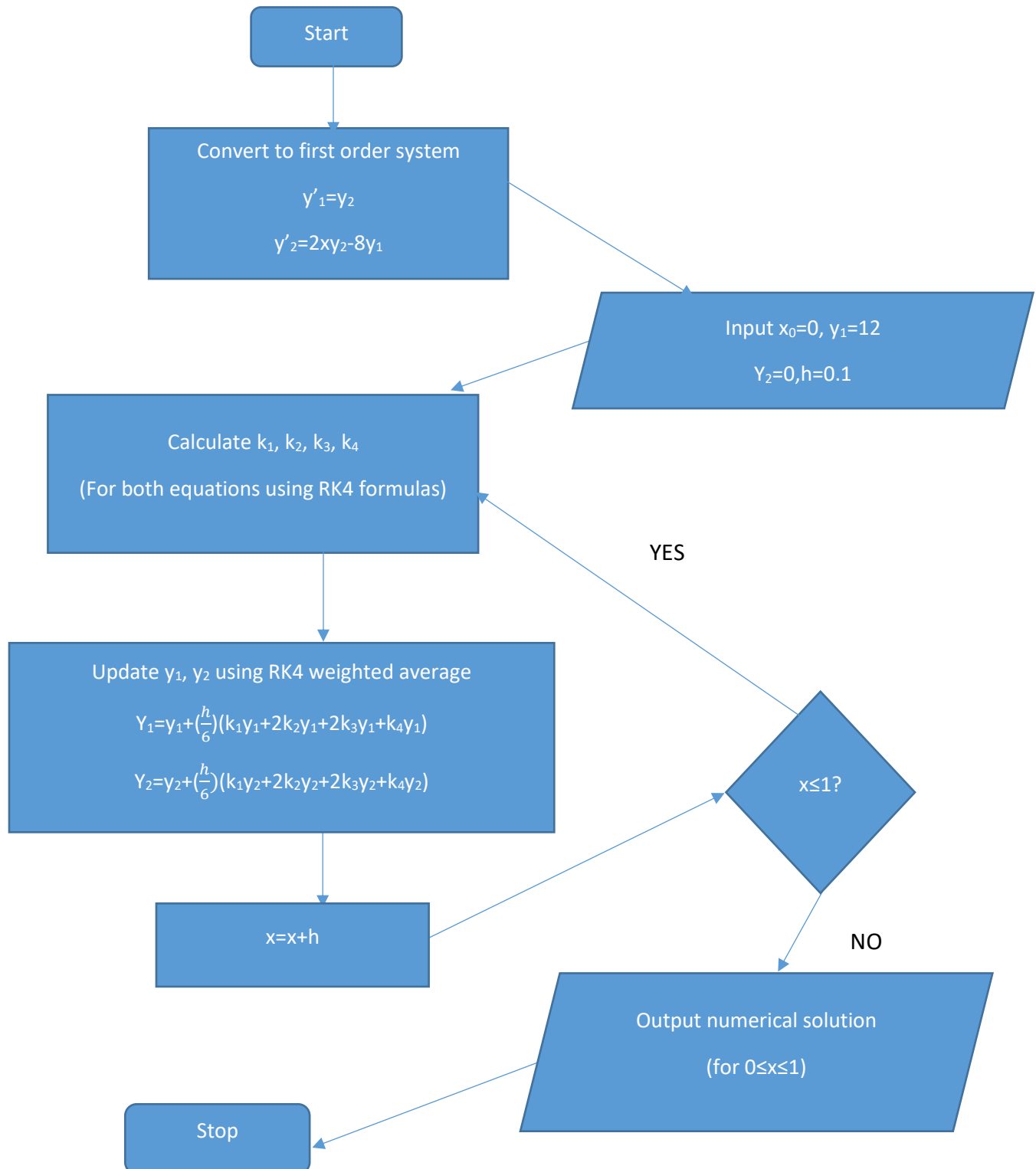


Use the fourth order Runge -Kutta alogarithm with step size 0.1 to find a numerical solution of the initial value problem;

$Y''-2xy'+8y=0$ with $y(0)=12$ and $y'(0)=0$ in the interval $0 \leq x \leq 1$ generate the assignment in word and power point files.



MATLAB CODE

```
%y'' - 2xy' + 8y = 0
%y(0)=12, y'(0)=0
h=0.1;
x=0;
y1=12;
y2=0;
N=10;
fprintf('x   y   y'\n');
fprintf('.....\n');
for i=1:N+1
    fprintf('%8.1f %12.6f %12.6f\n',x,y1,y2);
    k1_1=y2;
    k1_2=2*x*y2-8*y1;

    k2_1=y2+(h/2)*k1_2;
    k2_2=2*(x+h/2)*(y2+h/2)*k1_2; ... -8*(y1+(h/2)*k1_1);

    k3_1=y2+(h/2)*k2_2;
    k3_2=(x+h/2)*(y2+h/2*k2_2); ... -8*(y1+(h/2)*k2_1);

    k4_1=y2+h*k3_2;
    k4_2=2*(x+h)*(y2+h*k3_2); ... -8*(y1+h*k3_1);

    y1=y1+(h/6)*(k1_1 + 2*k2_1+2*k3_1+k4_1);
    y2=y2 + (h/6)*(k1_2 + 2*k2_2+2*k3_2+k4_2);
    x=x+h;
end
```

‘FOR’AND WHILE LOOPS

```
h=0.1;
x=0;
y1=12;
y2=0;
N=10;
fprintf('x   y   yprime\n');
fprintf('.....\n');
for i=1:N+1
    fprintf('%8.1f%12.6f%12.6f'\n',x,y1,y2);

    k1_1=y2;
    k1_2=2*x*y2-8*y1;

    k2_1=y2+(h/2)*k1_2;
```

```

k2_2=2*(x+h/2)*(y2+h/2)*k1_2;...-8*(y1+(h/2)*k1_1);

k3_1=y2+(h/2)*k2_2;
k3_2=(x+h/2)*(y2+h/2*k2_2);...-8*(y1+(h/2)*k2_1);

k4_1=y2+h*k3_2;
k4_2=2*(x+h)*(y2+h*k3_2);...-8*(y1+h*k3_1);

y1=y1+(h/6)*(k1_1 + 2*k2_1+2*k3_1+k4_1);
y2=y2 +(h/6)*(k1_2 + 2*k2_2+2*k3_2+k4_2);
x=x+h;
end

```

```

h=0.1;
x=0;
y1=12;
y2=0;
N=10;
fprintf('x  y yprime\n');
fprintf('.....\n');
while i<=N+1
    fprintf('%.1f%.12.6f%.12.6f'\n',x,y1,y2);

    k1_1=y2;
    k1_2=2*x*y2-8*y1;

    k2_1=y2+(h/2)*k1_2;
    k2_2=2*(x+h/2)*(y2+h/2)*k1_2;...-8*(y1+(h/2)*k1_1);

    k3_1=y2+(h/2)*k2_2;
    k3_2=(x+h/2)*(y2+h/2*k2_2);...-8*(y1+(h/2)*k2_1);

    k4_1=y2+h*k3_2;
    k4_2=2*(x+h)*(y2+h*k3_2);...-8*(y1+h*k3_1);

    y1=y1+(h/6)*(k1_1 + 2*k2_1+2*k3_1+k4_1);
    y2=y2 +(h/6)*(k1_2 + 2*k2_2+2*k3_2+k4_2);
    x=x+h;
end

```

PSEUDOCODE

START

Set $h=0.1$

Set $x=0$

Set $y_1=12$

Set $y_2=0$

WHILE $X \leq 1$

Calculate k_1 for y_1

Calculate k_1 for y_2

Calculate k_2 for y_1

Calculate k_2 for y_2

Calculate k_3 for y_1

Calculate k_3 for y_2

Calculate k_4 for y_1

Calculate k_4 for y_2

Update y_1 using RK4 formula

Update y_2 using RK4 formula

$X=x+h$

Display x, y_1, y_2

END WHILE

STOP